Operating Systems Lab

Lab 03 Report

Batch Scripting & Windows Automation

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Section / Semester	BSCS-5A
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Course	Operating Systems - Lab
Platform	Microsoft Windows 10/11 (Command Prompt, Notepad)

1. Objective

To understand Batch scripting in Windows and to create and execute .bat files that combine CMD commands, variables, user input, arithmetic, conditions (IF / ELSE, IF EXIST) and loops (FOR, FOR /L) for automating file and directory management.

2. Tools Used

Windows Command Prompt (cmd) and Notepad. All scripts were created as .bat files and run inside a dedicated practice folder (for example C:\OSLabPractice).

3. Procedure

1. Open Notepad (Win + R, type notepad, Enter).
2. Type the script, then File > Save As, set Save as type to All Files and save with the .bat extension.
3. Open Command Prompt, go to the practice folder with cd, and run the file by typing its name.
4. Compare the result with the expected output given for each practical.

4. Practicals

4.1 First Batch File (hello.bat)

File: hello.bat **Objective:** Create and run a first .bat file using echo and pause.

Script

```
echo Hello Students
pause
```

Explanation of Commands

Command	Explanation
echo Hello Students	Displays text on screen
pause	Waits for a key press so the window does not close

Expected Output

```
Hello Students
Press any key to continue . . .
```

4.2 Practical 1 - Automatically Create Folders

File: practical1_folders.bat **Objective:** Create multiple directories using one Batch file.

Script

```
@echo off

REM Create main folders
mkdir Lectures
mkdir Assignments
mkdir Backup

echo All folders created successfully.

pause
```

Explanation of Commands

Command	Explanation
@echo off	Hides the commands from output
REM ...	Comment, ignored during execution
mkdir	Creates a directory
echo	Displays confirmation
pause	Waits for a key press

Expected Output

```
All folders created successfully.
Press any key to continue . . .
```

4.3 Practical 2 - Create Folder Structure

File: practical2_structure.bat **Objective:** Create the Operating_Systems folder with four subfolders and display it with tree.

Script

```
@echo off

REM Create main directory
mkdir Operating_Systems

REM Enter main directory
cd Operating_Systems

REM Create subdirectories
mkdir Lectures
mkdir Assignments
mkdir Projects
mkdir Backup

REM Display structure
tree

pause
```

Explanation of Commands

Command	Explanation
cd Operating_Systems	Change Directory - moves into the main folder
mkdir ...	Creates the four subfolders

tree	Displays the folder structure graphically

Expected Output

```
Folder PATH listing
C:.
+---Assignments
+---Backup
+---Lectures
\---Projects
Press any key to continue . . .
```

4.4 Practical 3 - Student Information File

File: practical3_student_file.bat **Objective:** Create a text file using output redirection (> and >>).

Script

```
@echo off

echo Student Name: Ali > student.txt
echo Course: Operating Systems >> student.txt
echo Lab: Batch Scripting >> student.txt

type student.txt

pause
```

Explanation of Commands

Command	Explanation
>	Creates/overwrites the file with the output
>>	Appends output to the end of the file
type	Displays the file contents

Expected Output

```
Student Name: Ali
Course: Operating Systems
Lab: Batch Scripting
Press any key to continue . . .
```

4.5 Variables Example

File: variable_example.bat **Objective:** Store a value in a variable and display it using %name%.

Script

```
@echo off

REM Store student's name
set name=Ali

REM Display variable
echo Student Name: %name%

pause
```

Explanation of Commands

Command	Explanation
set name=Ali	Creates variable name with value Ali
%name%	Retrieves the value stored in the variable

Expected Output

```
Student Name: Ali
Press any key to continue . . .
```

4.6 Practical 4 - User Input

File: practical4_input.bat **Objective:** Accept input from the user with set /p.

Note: *Enter your own name when prompted.*

Script

```
@echo off

REM Ask user for name
set /p name=Enter your name:

REM Display personalized message
echo Welcome %name% to Operating Systems Lab.

pause
```

Explanation of Commands

Command	Explanation
set /p name=...	Prompts the user and stores the input in name
echo Welcome %name% ...	Displays a personalized message

Expected Output

```
Enter your name: Hussain
Welcome Hussain to Operating Systems Lab.
Press any key to continue . . .
```

4.7 Multiple Inputs

File: multiple_inputs.bat **Objective:** Take more than one input and display it using echo. (blank line).

Script

```
@echo off

set /p name=Enter your name:
set /p reg=Enter registration number:

echo.
echo Student Information
echo Name: %name%
echo Registration Number: %reg%

pause
```

Explanation of Commands

Command	Explanation
set /p	Takes name and registration number
echo.	Prints a blank line

Expected Output

```
Enter your name: Ali
Enter registration number: 2024-CS-01

Student Information
Name: Ali
Registration Number: 2024-CS-01
Press any key to continue . . .
```

4.8 Practical 5 - Simple Addition

File: practical5_addition.bat **Objective:** Perform arithmetic using set /a.

Script

```
@echo off

REM Take first number
set /p num1=Enter first number:

REM Take second number
set /p num2=Enter second number:

REM Add numbers
set /a result=num1+num2

REM Display result
echo Sum = %result%

pause
```

Explanation of Commands

Command	Explanation
set /p num1/num2	Stores the two numbers entered
set /a result=num1+num2	Performs integer addition
echo Sum = %result%	Displays the result

Expected Output

```
Enter first number: 10
Enter second number: 20
Sum = 30
Press any key to continue . . .
```

4.9 IF / ELSE - Age Eligibility

File: age_check.bat **Objective:** Make a decision using if / else and the GEQ operator.

Note: *Run twice: once with age 20 and once with age 15.*

Script

```
@echo off

set /p age=Enter your age:

if %age% GEQ 18 (
    echo Eligible
) else (
    echo Not Eligible
)

pause
```

Explanation of Commands

Command	Explanation
if %age% GEQ 18	Checks whether age is greater than or equal to 18
else	Runs when the condition is false

Expected Output

```
Enter your age: 20
Eligible
Press any key to continue . . .
```

4.10 Practical 6 - Pass/Fail Script

File: practical6_passfail.bat **Objective:** Display PASS or FAIL depending on marks.

Script

```
@echo off

REM Ask student marks
set /p marks=Enter marks:

REM Check result
if %marks% GEQ 50 (
   echo Result: PASS
) else (
   echo Result: FAIL
)

pause
```

Explanation of Commands

Command	Explanation
set /p marks	Reads the marks
if %marks% GEQ 50	PASS if marks are 50 or more, else FAIL

Expected Output

```
Enter marks: 65
Result: PASS
Press any key to continue . . .
```

4.11 File Existence Check (IF EXIST)

File: exist_check.bat **Objective:** Check whether a file exists before acting.

Note: *Run once without notes.txt, then create notes.txt and run again.*

Script

```
@echo off

if exist notes.txt (
   echo File found.
) else (
   echo File does not exist.
)

pause
```

Explanation of Commands

Command	Explanation
if exist notes.txt	Checks if notes.txt exists in the current folder

Expected Output

```
File does not exist.   (or "File found." if notes.txt is present)
```

4.12 Backup Only If File Exists

File: backup_if_exists.bat **Objective:** Combine if exist, if not exist and copy for a safe backup.

Note: *Create report.txt in the same folder first.*

Script

```
@echo off

REM Check if file exists
if exist report.txt (
```

```
    REM Create backup directory if needed
    if not exist Backup mkdir Backup

    REM Copy file
    copy report.txt Backup

    echo Backup completed.

) else (

    echo Error: report.txt was not found.

)

pause
```

Explanation of Commands

Command	Explanation
if exist report.txt	Proceeds only if the report exists
if not exist Backup mkdir Backup	Creates Backup only if missing
copy report.txt Backup	Copies the file into Backup

Expected Output

```
        1 file(s) copied.
Backup completed.
Press any key to continue . . .
```

4.13 Practical 7 - Create Folders Using a Loop

File: practical7_loop_folders.bat **Objective:** Use a for loop to create Student1 to Student5.

Script

```
@echo off

for %%i in (1 2 3 4 5) do (
    mkdir Student%%i
)

echo Folders created successfully.

pause
```

Explanation of Commands

Command	Explanation
for %%i in (1 2 3 4 5)	Loops over the values 1 to 5
mkdir Student%%i	Creates Student1 ... Student5

Expected Output

```
Folders created successfully.
Press any key to continue . . .
```

4.14 FOR /L Loop

File: forL.bat **Objective:** Use a numeric for /L loop to print numbers and create Lab1 to Lab10.

Script

```
@echo off

for /L %%i in (1,1,5) do echo %%i

for /L %%i in (1,1,10) do (
   mkdir Lab%%i
)

echo Lab folders created.

pause
```

Explanation of Commands

Command	Explanation
for /L %%i in (1,1,5)	Counts from 1 to 5 with step 1
mkdir Lab%%i	Creates Lab1 ... Lab10

Expected Output

```
1
2
3
4
5
Lab folders created.
Press any key to continue . . .
```

4.15 Window Customization (title and color)

File: title_color.bat **Objective:** Change the CMD window title and colors.

Script

```
@echo off

title OS Lab Automation
color 0A

echo Title and color changed.

pause
```

Explanation of Commands

Command	Explanation
title OS Lab Automation	Sets the window title
color 0A	Black background, light green text

Expected Output

```
Title and color changed.
Press any key to continue . . .  (green text, new window title)
```

4.16 Practical 8 - Complete Batch Automation

File: practical8_automation.bat **Objective:** Automate folder creation, file creation, backup and display in one script.

Script

```
@echo off

REM Set CMD window title
title OS Lab Student Automation

REM Ask for student name
set /p name=Enter student name:

REM Create main folder
mkdir Student_Record

REM Enter folder
cd Student_Record

REM Create student information file
echo Student Name: %name% > student.txt
echo Course: Operating Systems >> student.txt
echo Lab: Batch Scripting >> student.txt

REM Create backup folder if it does not exist
if not exist Backup mkdir Backup

REM Copy information file into backup
copy student.txt Backup

REM Display contents
echo.
echo Student Record:
type student.txt

echo.
echo Automation completed successfully.

pause
```

Explanation of Commands

Command	Explanation
title	Sets window title
set /p name	Accepts the student name
mkdir / cd	Creates and enters Student_Record
echo > / >>	Writes student.txt
if not exist Backup mkdir Backup	Creates Backup if missing
copy student.txt Backup	Creates backup copy
type student.txt	Displays file contents

Expected Output

```
Enter student name: Ali
        1 file(s) copied.

Student Record:
Student Name: Ali
Course: Operating Systems
```

```
Lab: Batch Scripting

Automation completed successfully.
Press any key to continue . . .
```

4.17 Student Hands-On Activity - Course Workspace Creator

File: course_workspace.bat **Objective:** Create a course workspace with subfolders, an info file, a backup copy and the directory tree.

Script

```
@echo off

REM Ask user for course name
set /p course=Enter course name:

REM Create main course directory
mkdir "%course%"

REM Move into main directory
cd "%course%"

REM Create subdirectories
mkdir Lectures
mkdir Labs
mkdir Assignments
mkdir Backup

REM Create course information
echo Course Name: %course% > course_info.txt

REM Copy information file to backup
copy course_info.txt Backup

REM Display directory structure
tree /f

REM Display completion message
echo.
echo Course workspace created successfully.

pause
```

Explanation of Commands

Command	Explanation
mkdir "%course%"	Quotes allow folder names with spaces
tree /f	Shows folders and files
copy	Backs up course_info.txt

Expected Output

```
Enter course name: Operating Systems
        1 file(s) copied.
(tree listing of Lectures, Labs, Assignments, Backup and course_info.txt)

Course workspace created successfully.
Press any key to continue . . .
```

4.18 Basic Example 16.1 - Three Folders

File: worked_16_1_three_folders.bat **Objective:** Create folders Theory, Lab and Assignment.

Note: *Run in a fresh folder to avoid "already exists" errors.*

Script

```
@echo off

mkdir Theory
mkdir Lab
mkdir Assignment

pause
```

Explanation of Commands

Command	Explanation
mkdir	Creates each of the three directories

Expected Output

```
Press any key to continue . . .
```

4.19 Intermediate Example 16.2 - Name and Registration Number

File: worked_16_2_name_reg.bat **Objective:** Ask name and registration number and save both in a file.

Script

```
@echo off

set /p name=Enter Name:
set /p reg=Enter Registration Number:

echo Name: %name% > student.txt
echo Registration Number: %reg% >> student.txt

type student.txt

pause
```

Explanation of Commands

Command	Explanation
set /p	Takes name and registration number
> / >>	Writes both lines into student.txt
type	Displays the file

Expected Output

```
Enter Name: Ali
Enter Registration Number: 2024-CS-01
Name: Ali
Registration Number: 2024-CS-01
Press any key to continue . . .
```

4.20 Problem-Solving Example 16.3 - Report Backup

File: worked_16_3_report_backup.bat **Objective:** Back up report.txt only if it exists.

Note: *Run once without report.txt and once with it.*

Script

```
@echo off

if exist report.txt (

    if not exist Backup mkdir Backup

    copy report.txt Backup

    echo Report backup completed.

) else (

    echo ERROR: report.txt does not exist.

)

pause
```

Explanation of Commands

Command	Explanation
if exist report.txt	Checks the condition first
copy report.txt Backup	Copies only when the file exists

Expected Output

```
        1 file(s) copied.
Report backup completed.   (or: ERROR: report.txt does not exist.)
```

4.21 Question 1 - Welcome Message

File: q1_welcome.bat **Objective:** Print Welcome to Operating Systems Lab.

Script

```
@echo off
echo Welcome to Operating Systems Lab
pause
```

Explanation of Commands

Command	Explanation
echo	Displays the message

Expected Output

```
Welcome to Operating Systems Lab
Press any key to continue . . .
```

4.22 Question 2 - Windows, Linux, Backup Folders

File: q2_three_directories.bat **Objective:** Create directories Windows, Linux and Backup.

Note: *Run in a fresh folder.*

Script

```
@echo off

mkdir Windows
mkdir Linux
mkdir Backup

pause
```

Explanation of Commands

Command	Explanation
mkdir	Creates the three directories

Expected Output

```
Press any key to continue . . .
```

4.23 Question 3 - Create Folder from User Input

File: q3_folder_input.bat **Objective:** Ask for a folder name and create it.

Script

```
@echo off

set /p folder=Enter folder name:

mkdir "%folder%"

echo Folder created successfully.

pause
```

Explanation of Commands

Command	Explanation
set /p folder	Reads the folder name
mkdir "%folder%"	Quotes handle names with spaces, e.g. Operating Systems

Expected Output

```
Enter folder name: Operating Systems
Folder created successfully.
Press any key to continue . . .
```

4.24 Question 4 - Check assignment.txt

File: q4_assignment_check.bat **Objective:** Determine whether assignment.txt exists.

Note: *Run with and without assignment.txt.*

Script

```
@echo off

if exist assignment.txt (
    echo Assignment file found.
) else (
    echo Assignment file not found.
)

pause
```

Explanation of Commands

Command	Explanation
if exist assignment.txt	Tests for the file
else	Runs when the file is missing

Expected Output

```
Assignment file not found.   (or "Assignment file found.")
```

4.25 Question 5 - Folders Week1 to Week10

File: q5_week_folders.bat **Objective:** Create Week1 ... Week10 automatically.

Script

```
@echo off

for /L %%i in (1,1,10) do (
    mkdir Week%%i
)

echo All folders created.

pause
```

Explanation of Commands

Command	Explanation
for /L %%i in (1,1,10)	Loops 1 to 10
mkdir Week%%i	Creates Week1 ... Week10

Expected Output

```
All folders created.
Press any key to continue . . .
```

4.26 Basic Script Structure (Section 6.4) and Notes File (Section 8.1)

File: structure_and_notes.bat **Objective:** Show REM comments, mkdir and echo redirection together.

Script

```
@echo off

REM Display heading
echo Operating Systems Lab

REM Create folder
mkdir OS_Lab

REM Create a notes file
echo Operating Systems Lab > notes.txt

REM Display completion message
echo Folder created successfully.

pause
```

Explanation of Commands

Command	Explanation
REM	Comment lines
mkdir OS_Lab	Creates folder
echo ... > notes.txt	Creates notes.txt

Expected Output

```
Operating Systems Lab
Folder created successfully.
Press any key to continue . . .
```

4.27 Instructor Demo - 20 Student Folders

File: demo_20_folders.bat **Objective:** Create Student1 ... Student20 with one loop.

Script

```
@echo off

for /L %%i in (1,1,20) do mkdir Student%%i

echo 20 folders created.

pause
```

Explanation of Commands

Command	Explanation
for /L %%i in (1,1,20)	Repeats mkdir 20 times

Expected Output

```
20 folders created.
Press any key to continue . . .
```

4.28 Lab Task - Automatic Student Record Generator

File: student_record.bat **Objective:** Build the final submission script: collect student details, create folder and file, back up and verify.

Script

```
@echo off
title Student Record Generator

REM Take student details
set /p name=Enter student name:
set /p reg=Enter registration number:
set /p sem=Enter semester:

REM Create folder named after registration number
if not exist "%reg%" mkdir "%reg%"
cd "%reg%"

REM Create student.txt and save details
echo Student Name: %name% > student.txt
echo Registration Number: %reg% >> student.txt
echo Semester: %sem% >> student.txt

REM Create Backup folder and copy file
if not exist Backup mkdir Backup
copy student.txt Backup

REM Check that the backup exists
if exist Backup\student.txt (
   echo Backup Successful
) else (
   echo Backup Failed
)

REM Display complete directory structure
echo.
tree /f

pause
```

Explanation of Commands

Command	Explanation
set /p (x3)	Reads name, registration number and semester
mkdir "%reg%" / cd	Creates and enters the folder named after reg. no.
echo > / >>	Saves details in student.txt
if not exist Backup mkdir Backup	Creates Backup folder
copy student.txt Backup	Copies the file into Backup
if exist Backup\student.txt	Verifies backup, prints Backup Successful
tree /f	Displays the complete directory structure
pause	Pauses before exit

Expected Output

```
Enter student name: Ali
Enter registration number: 2024-CS-01
Enter semester: 3
        1 file(s) copied.
Backup Successful
```

```
(tree /f listing: Backup\student.txt and student.txt)
Press any key to continue . . .
```

5. Short Questions

Question	Answer
What is a Batch file?	A text file containing Windows commands that execute sequentially.
What does @echo off do?	Hides command lines from normal output.
Difference between > and >>?	> creates/overwrites a file; >> appends to it.
What is set /p used for?	To accept input from the user.
What does IF EXIST do?	Checks whether a file or directory exists.
Why are loops useful?	They repeat operations automatically, e.g. creating many folders.

6. Conclusion

In this lab, manual CMD commands were converted into Batch scripts. Variables, user input, arithmetic, IF / ELSE conditions, file-existence checks and FOR loops were used to automate folder creation, file writing and backups. The final student_record.bat script combined all these concepts into one automated workflow, showing how Batch scripting saves time and makes repetitive tasks consistent.